

INVESTMENT COMMITTEE Q&A PREP SHEET

Objection Defense, Geopolitical Risk Mitigation & Hardware Bottleneck Verification

1. EXECUTIVE SUMMARY & DEFENSE STRATEGY

This preparation memorandum anticipates the primary objections and risk concerns expected from the Investment Committee regarding the proposed **Humanoid Actuator Basket** (Tuopu 30%, Sanhua 25%, Inovance 20%, HL Mando/Moog 15%, Harmonic Drive 10%). The thesis centers on backing **Quadrant 1 Tier-1 manufacturing backstops** rather than pre-IPO robot OEMs. Core pushback areas include China A-share geopolitical exposure, rare-earth magnet export controls, machining yields, and single-OEM timeline dependency.

Risk Category	Committee Pushback / Objection	Core Defense & Structural Mitigation
Geopolitical Risk & China A-Shares (Tuopu, Sanhua, Inovance)	Over 75% of portfolio exposure is tied to Chinese Tier-1 suppliers subject to delisting, cross-border capital controls, or tariff escalations.	• Dual-Sourcing & Global Plant Footprint: Both Tuopu and Sanhua operate operational manufacturing campuses in North America (Mexico / US) to bypass direct tariffs. • Stock Connect Liquidity: Access via HKEX Northbound Connect provides daily liquidity with no lock-up periods. • Western Hedge Basket: 25% allocation into US defense (Moog), Korean auto (HL Mando), and Japanese precision (Harmonic Drive) protects against unilateral trade freezes.
Rare-Earth Permanent Magnets (NdFeB / Dysprosium Constraints)	China controls over 85% of global rare-earth refining. Export restrictions on NdFeB magnets could paralyze frameless motor production for linear/rotary actuators.	• Grain Boundary Diffusion (GBD) Tech: Sanhua and Tuopu utilize GBD manufacturing, reducing heavy rare-earth (Dysprosium/Terbium) content by 50%–70% while maintaining thermal torque density. • Pass-Through Pricing Contracts: Tier-1 OEM agreements incorporate raw-material index clause pass-throughs, protecting supplier gross margins against commodity spikes.
Machining Bottlenecks & Yields (Planetary Roller Screws)	Sub-micron grinding tolerances for inverted planetary roller screws currently yield only ~50% in pilot lines, creating severe revenue delivery bottlenecks.	• Pilot Billing Mechanics: Tier-1 suppliers bill OEMs on component delivery during factory pilot ramps, ensuring revenue capture prior to full end-consumer robot deployment. • Yield Progression Trajectory: Automated multi-axis grinding lines are projected to scale yields from 50% to 85%+ over Years 1–2, expanding gross margins as scrap rates plummet.
Single-OEM Dependency (Tesla Optimus V3 Timeline)	Performance depends heavily on Tesla's Optimus V3 pilot execution. Delays in mass deployment directly impact supplier top-line volume.	• Cross-Platform Applicability: Rotary/linear actuators and servo drives from Inovance and Sanhua are platform-agnostic, supplying competing Chinese and Western humanoid OEMs (Unitree, AgiBot, Boston Dynamics). • Core Automotive Base: Tuopu and Sanhua maintain profitable legacy automotive Tier-1 revenue baselines, insulating downside risk.

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Margin Compression & Scaling (ASP \$15k to \$4.5k Reduction)	Actuator set ASPs drop from \$15,000 to \$4,500 over 5 years. Margin compression could erode expected portfolio returns.	• Operating Leverage at Scale: Fixed cost absorption at 1,000,000 units/year scale offsets unit price reductions. • 32% Gross Margin Buffer: Proprietary precision machining tolerances protect Tier-1 gross margins well above standard 10–15% automotive commodity tiers.

SUMMARY DEFENSE STATEMENT FOR COMMITTEE CHAIR

'We are not taking speculative pre-IPO startup risk. By positioning capital in Quadrant 1 automotive Tier-1 leads, we back companies with established, cash-generative core businesses that hold locked-in positions on the humanoid Bill of Materials. Our dual-sourcing footprint across Mexico and Asia, combined with pass-through raw material contracts and cross-platform hardware applicability, insulates our capital while capturing a 36.5% CAGR as factories scale from 50% pilot yield to 1 million units annual capacity.'